C2 COMPONENT ARCHITECTURE

Endpoint → Domain Service → Repository / SQL Outbox; sáu miền song song, không truy vấn chéo database.

MEMBER

ADMIN

REACT SPA / NGINX

HTTP / JWT

YARP — AUTHENTICATED ROUTES

REST route dispatch

IDENTITY — REST :8081

REST

ENDPOINT / CONTROLLER
Auth / Profile / JWT

Validate / business rules

APPLICATION / DOMAIN
Xác thực, phân quyền
Refresh Token, hồ sơ

Transaction / EF Core

REPOSITORY / DbContext
LongevityIdentityDb / IdentityOutbox

Outbox

CATALOG — REST :8082

REST

ENDPOINT / CONTROLLER
Foods / Recipes / Rules

Validate / business rules

APPLICATION / DOMAIN
Quy tắc dinh dưỡng
Duyệt và lưu phiên bản

Transaction / EF Core

REPOSITORY / DbContext
LongevityCatalogDb / Rule Audits / Events

Domain event

PLANNING — REST :8083

REST

ENDPOINT / CONTROLLER
Meal Plan / FMD / Challenge

Validate / business rules

APPLICATION / DOMAIN
Điều phối thực đơn
Cổng an toàn FMD

Transaction / EF Core

REPOSITORY / DbContext
LongevityPlanningDb / PlanningOutbox

Outbox

TRACKING — REST :8084

REST

ENDPOINT / CONTROLLER
Meal Logs / Activity / LDAS

Validate / business rules

APPLICATION / DOMAIN
Nhật ký, cửa sổ ăn
Chấm điểm LDAS

Transaction / EF Core

REPOSITORY / DbContext
LongevityTrackingDb / TrackingOutbox

Outbox

RECOMMENDATION — REST :8087 / gRPC :8085

REST

ENDPOINT / CONTROLLER
Gợi ý món / Phản hồi

Validate / business rules

APPLICATION / DOMAIN
Lọc dị ứng (hard constraint)
Xếp hạng theo RuleVersion

Transaction / EF Core

REPOSITORY / DbContext
LongevityRecommendationDb / RecommendationOutbox

Outbox

WORKER / OPS — REST :8086 + Worker

REST

ENDPOINT / CONTROLLER
Job / Event / Dead-letter

Validate / business rules

APPLICATION / DOMAIN
Nhắc lịch / Báo cáo
Consumer, retry và audit

Transaction / EF Core

REPOSITORY / DbContext
LongevityWorkerDb / WorkerOutbox / Inbox

Outbox

Events

OUTBOX EVENT DELIVERY
Redis publish — Replay — Inbox / DLQ

Primary

REDIS STREAMS — ACTIVE
XADD / XREADGROUP / XACK

Audit mirror

RABBITMQ — OPTIONAL QA
Exchange / Durable queue / ACK

C2 là mô hình thành phần mục tiêu: Redis hoạt động, RabbitMQ hỗ trợ audit mirror khi bật. Một số API legacy chưa được cutover.